

QRAOS™ Mobile PWA

Master User, Deployment, Operations and Developer Handoff Guide

GitHub Pages Audited Proof-of-Concept Build

Version: Free Local-First Edition

Prepared: July 2026

Deployment target: GitHub Pages (main branch / repository root)

Purpose of this guide

This is the complete operating manual for deploying, using, testing, backing up, updating and handing off the QRAOS proof of concept to developers. It documents the current build as it exists; it does not claim that production-only functions are already present.

CONFIDENTIAL WORKING GUIDE - PROOF OF CONCEPT

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Navigation note

The Word version uses structured headings so developers can use Word's Navigation pane. Page numbering may update when opened in Microsoft Word.

1. Executive Overview

QRAOS™ Mobile is a static, local-first Progressive Web Application (PWA) designed to screen and organize real estate development opportunities. The proof of concept runs entirely in the browser and can be published free on GitHub Pages.

Area	Current behavior
Hosting	Static files hosted on GitHub Pages over HTTPS.
Cost	No server, database, AI API or App Store fee is required for the proof of concept.
Data storage	Project records are stored in the browser's localStorage on the device being used.
Artificial intelligence	The app generates a structured package that the user manually copies into ChatGPT; the response can be pasted back into the project.
Offline behavior	The application shell and core files are cached by a service worker. Previously saved browser data remains local.
Best use	Initial screening, structured due diligence, preliminary financial feasibility and professional briefing.
Not production-ready for	Secure collaboration, confidential document storage, automatic municipal verification or full lender/CMHC underwriting.

Core operating principle

Use QRAOS to decide whether a property deserves more time and professional expense. Treat all zoning, legal, engineering, environmental and lending outputs as preliminary until independently verified.

1.1 Intended users

- Owner or acquisition lead screening new opportunities.
- Employee or technologist collecting property and municipal information.
- Architect or urban planner receiving a structured feasibility brief.
- Broker, lawyer or notary reviewing draft conditions and identified risks.
- Developer or software team evaluating the proof of concept for a production rebuild.

1.2 Expected outcome for each property

- A saved project record and pipeline stage.
- Basic property and zoning assumptions.
- Preliminary buildable area and unit-yield estimates.
- Construction cost and residual land value screening.
- Financing sensitivity.
- AI due-diligence package and imported analysis.
- Municipal question list, architect brief and draft offer conditions.
- A printable professional summary report.

- A JSON backup file.

2. Current Capabilities and Limitations

2.1 Functions included

Module	What it currently does
Dashboard	Displays the selected project, saved projects and next actions.
Property & Pipeline	Captures project name, address, municipality, stage, price, units, lot details, listing link and notes.
Zoning & Feasibility	Captures zoning assumptions and calculates preliminary footprint, gross floor area, unit yield and parking.
Scenario Comparison	Compares alternative unit counts against the entered development assumptions.
Construction Cost Engine	Combines hard cost, demolition, site work, fees, soft costs, contingency, carrying cost, taxes and other costs.
Residual Land Value	Estimates income, NOI, stabilized value, project cost, residual land value and preliminary maximum offer.
CMHC / ACLP Sensitivity	Provides preliminary LTC/DSCR-based loan and equity sensitivity.
AI Due Diligence	Creates a structured copy/paste package for ChatGPT and stores the returned analysis.
Municipal Intelligence	Stores verified facts and assumptions and generates a municipal question list.
Architect AI	Generates a preliminary architect feasibility brief.
Offer Generator	Produces selected draft due-diligence conditions for professional review.
Knowledge Base	Provides a starter searchable library of due-diligence and clause concepts.
Professional Report	Builds a printable project summary that can be saved as PDF.
Export	Downloads the active project as a JSON backup.

2.2 Functions not included

Not included	Practical consequence
Automatic Centris/MLS/REALTOR.ca import	Listing information must be entered or copied manually.
Municipal database integration	Zoning facts must be researched and verified by the user or professionals.
Cloud database	Projects do not automatically synchronize between devices.
User login or permissions	There is no account, role, password or team access control.

Not included	Practical consequence
Secure document repository	Do not upload sensitive documents to the public GitHub repository.
Direct ChatGPT integration	AI analysis requires manual copy and paste.
Complete CMHC underwriting	The financing module is a screening sensitivity, not lender approval.
Full pro forma	No detailed monthly cash flow, IRR, taxes, draws, waterfalls or investor accounting.
Professional certification	Outputs are not legal, architectural, engineering, appraisal or lending opinions.
Automatic restore button	The current proof of concept exports JSON; restoration may require a future import function or developer assistance.

3. System Architecture and Data Behavior

```

GitHub Pages (public static host)
  |
  v
index.html + manifest.webmanifest + service-worker.js + icons
  |
  v
Browser application (Safari / Chrome / Edge)
  |
  +--> localStorage: project records on this device
  +--> JSON export: user-created backup file
  +--> ChatGPT: manual copy/paste workflow
  +--> OpenStreetMap: user-opened external map search

```

3.1 Files in the deployment package

File	Purpose	Required
index.html	Complete application interface, calculations and browser logic.	Yes
manifest.webmanifest	PWA name, icons, start URL, display mode and scope.	Yes
service-worker.js	Offline caching and update behavior.	Yes
icon-192.png	Home Screen and PWA icon.	Yes
icon-512.png	High-resolution PWA icon.	Yes
.nojekyll	Prevents unnecessary Jekyll processing on GitHub Pages.	Recommended

File	Purpose	Required
README.md	General package information.	No
GITHUB_PAGES_SETUP.md	Condensed deployment directions.	No
AUDIT_REPORT.json	Compatibility audit result.	No
START_WINDOWS.bat / START_MAC.command	Local computer launch helpers; not used by GitHub Pages.	No

Important data fact

Uploading the application files does not upload the projects you later enter. The projects are stored in the browser on each device. A visitor to the public URL opens a separate blank local workspace.

3.2 Local storage behavior

- An iPhone project remains on that iPhone and in that browser profile.
- A computer project does not automatically appear on the iPhone.
- Clearing Safari or browser website data can erase saved projects.
- Private browsing is unsuitable because storage may not persist.
- Export the active project regularly and retain the JSON file in a controlled folder.

4. GitHub Pages Deployment - Complete First-Time Setup

Use the audited package

Deploy the files from the audited ZIP, not an earlier prototype package. The repository must contain index.html directly at its root.

4.1 Before starting

- ☐ A GitHub account.
- ☐ The extracted audited QRAOS ZIP folder.
- ☐ A modern browser on a computer.
- ☐ An iPhone or iPad for installation testing.
- ☐ A non-confidential test property.

4.2 Create the repository

1. Sign in to GitHub

Open GitHub in your browser and sign in. Create a free account first if required.

2. Start a new repository

Use the plus sign in the upper-right corner and select New repository.

3. Name the repository

Use a simple lowercase name such as qraos-mobile. The name becomes part of the published URL.

4. Choose visibility

Select Public for the simplest free GitHub Pages proof-of-concept deployment.

5. Do not initialize extra files

Leave Add a README, .gitignore and license unchecked because the package already contains supporting files.

6. Create the repository

Click Create repository.

Expected repository URL:
`https://github.com/YOUR-USERNAME/qraos-mobile`

4.3 Extract and upload the application files

1. Extract the ZIP

On Windows, right-click the ZIP and select Extract All. On Mac, double-click it.

2. Open the extracted folder

Confirm that index.html, manifest.webmanifest and service-worker.js are visible.

3. Open the empty GitHub repository

Choose uploading an existing file or Add file > Upload files.

4. Upload individual files

Drag the extracted files into GitHub. Do not upload only the ZIP.

5. Confirm root placement

index.html must be visible on the repository's first page, not inside an extra folder.

6. Commit the upload

Use a message such as Initial QRAOS deployment and commit directly to main.

Correct root structure	Incorrect nested structure
graos-mobile/index.html	graos-mobile/QRAOS-folder/index.html
graos-mobile/manifest.webmanifest	graos-mobile/QRAOS-folder/manifest.webmanifest
graos-mobile/service-worker.js	graos-mobile/QRAOS-folder/service-worker.js

4.4 Enable GitHub Pages

1. Open repository Settings

Select Settings at the top of the repository.

2. Open Pages

In the left menu under Code and automation, select Pages.

3. Choose deployment source

Set Source to Deploy from a branch.

4. Select the branch

Choose main.

5. Select the folder

Choose / (root).

6. Save

Click Save and wait for deployment.

7. Find the published address

Refresh the Pages screen until GitHub shows the live HTTPS URL.

Expected public app URL:
https://YOUR-USERNAME.github.io/graos-mobile/

Deployment delay

A first deployment or update can take several minutes. A temporary 404 immediately after saving does not necessarily mean the setup failed.

4.5 Verify deployment

- ☐ The QRAOS dashboard opens instead of a 404 page.
- ☐ The URL begins with https://.
- ☐ The browser console is not showing a fatal JavaScript error.
- ☐ New Project and Save respond.
- ☐ Navigation buttons open Dashboard, Property, Finance, Tools and Report.

- A saved test project remains after closing and reopening the page.
- The manifest and service worker files return successfully when opened directly.

5. Install on iPhone, iPad and Desktop

5.1 iPhone or iPad

1. Open Safari

Use Safari, not an in-app browser opened from email or social media.

2. Open the GitHub Pages URL

Enter the complete HTTPS address including the repository name.

3. Allow the app to load

Wait until the dashboard appears.

4. Open Share

Tap the Share icon.

5. Add to Home Screen

Scroll and choose Add to Home Screen.

6. Name the app

Use QRAOS or QRAOS Mobile.

7. Confirm installation

Tap Add, then launch the new icon from the Home Screen.

iPhone data location

The Home Screen app and Safari use the website's browser storage. Do not clear its website data without exporting important projects first.

5.2 Desktop

- Open the same GitHub Pages URL in Chrome, Edge or Safari.
- Chrome and Edge may display an Install icon in the address bar.
- Desktop data is separate from iPhone data unless a future cloud backend is added.
- Use desktop for extensive typing, calculations and report printing.

6. First Launch and First Test Project

6.1 Safe first test

Do not begin with a confidential live deal

First confirm the storage, calculations, report and export workflow using fictional data.

Field	Suggested test value
Project name	Test 20-Unit Project
Address	123 Test Street
Municipality	Montréal
Stage	Initial Screen
Asking price	1,200,000
Existing units	1
Target units	20
Lot area	10,000 sq ft
Frontage	100 ft
Depth	100 ft

6.2 First-launch procedure

1. Tap New

Create a blank project.

2. Enter identification fields

Complete the project name, address, municipality and stage.

3. Enter property facts

Add price, units, lot dimensions, listing link and notes.

4. Tap Save

Save before moving to another module.

5. Close and reopen the app

Confirm that the project remains in the Projects list.

6. Run one calculation

Enter zoning and financial assumptions and confirm outputs change.

7. Generate one AI package

Confirm the text package can be copied.

8. Build the report

Review the summary and use Print / Save PDF.

9. Export the project

Save the JSON file to a known folder.

6.3 Expected first-test result

- ☐ Project appears in the Projects list.
- ☐ Project remains after the application is closed and reopened.
- ☐ Calculated values update when assumptions change.
- ☐ Generated prompts and drafts contain the entered project data.
- ☐ The report is printable.
- ☐ A JSON backup downloads successfully.

7. Complete Module-by-Module User Guide

7.1 Dashboard

Control	How to use it
Current Project	Confirms which project is active. Always verify this before editing.
New	Starts a blank property record. Save or export the current project first if needed.
Save	Stores the active project in local browser storage.
Export	Downloads the active project as JSON for backup or developer review.
Projects	Lists saved projects. Select one to make it active.
Next Actions	Shows reminders or generated action items associated with the current workflow.
Bottom navigation	Moves among Dashboard, Property, Finance, Tools and Report.

7.2 Property & Pipeline

Field	What to enter	Control rule
Project name	Short internal deal name.	Make it unique and recognizable.
Address	Full civic address.	Do not rely on address alone for legal lot identification.
Municipality	Municipality or borough.	Use the authority that controls zoning.
Stage	Current acquisition stage.	Update whenever the deal advances or is rejected.
Asking price	Seller's current asking price.	Use digits without unnecessary text.
Existing units	Current known dwelling count.	Mark uncertainty in notes if not legally verified.

Field	What to enter	Control rule
Target units	Desired development scenario.	Treat as a target, not an entitlement.
Lot area/frontage/depth	Known site dimensions.	Use the certificate of location or verified records when available.
Listing URL	Source link.	Links may expire; retain key facts in notes.
Listing notes	Broker claims, condition, opportunities and risks.	Separate statements from verified facts.
Open free map	Opens an external map search.	Mapping does not confirm cadastral or zoning boundaries.

7.3 Zoning & Feasibility

Input	Meaning	Recommended source
Zone	Municipal zoning designation.	Official zoning map or written municipal confirmation.
Multifamily	Whether multifamily use appears permitted.	Applicable bylaw and municipal planner.
Max units	Maximum dwellings assumed.	Density rules and site-specific verification.
Max storeys	Assumed storey limit.	Zoning, height rules and design constraints.
Coverage %	Maximum building footprint as a percentage of lot area.	Applicable zoning standard.
FAR / FSI	Maximum floor area relative to lot area.	Applicable zoning standard.
Setbacks	Required front, rear and side clearances.	Zoning and existing non-conformity review.
Avg unit size	Average gross or net unit assumption used by the model.	Concept program.
Efficiency %	Share of gross area usable for units.	Architect's preliminary test fit.
Parking/unit	Required or assumed parking ratio.	Bylaw and exemptions.
Approval path	As-of-right, variance, PIIA, PPCMOI or other path.	Municipal planning review.
Servicing	Water, sewer and utility status.	Municipality and engineers.
Fatal issue	Known condition that could invalidate the concept.	Written evidence and professional review.

Avoid false precision

A precise-looking unit count can still be wrong when the zoning, setbacks, circulation, fire code, parking, servicing

or efficiency assumptions are incomplete.

7.4 Scenario Comparison

Use this screen to compare alternative unit counts under the same base assumptions. The comparison helps identify a plausible range before engaging an architect.

- Start with conservative, base and upside scenarios.
- Compare area required per scenario against the preliminary maximum gross floor area.
- Reject scenarios that depend on unverified discretionary approvals unless clearly labelled.
- Carry the best two or three scenarios into the architect brief.

7.5 Construction Cost Engine

Input	Use
Hard cost / gross sq ft	Construction cost multiplied by gross building area.
Demolition	Building removal, abatement and related work.
Site work	Excavation, retaining, landscaping, civil and exterior work.
Municipal & utility fees	Permits, development charges, connections and utility costs.
Professional fees %	Architect, engineering, planning, survey and consultant allowance.
Other soft costs %	Legal, insurance, marketing, project administration and similar costs.
Contingency %	Allowance for uncertainty and unforeseen work.
Financing & carrying	Interest, lender fees, taxes, insurance and holding costs.
Taxes not recoverable	Non-recoverable GST/QST or similar tax assumptions.
Other costs	Items not captured elsewhere.

- Use gross floor area consistently.
- Do not mix cost-per-net-rentable-square-foot with gross area.
- Document the date and source of each cost assumption in project notes.
- Run conservative and stressed cost cases before setting a maximum offer.

7.6 Residual Land Value

Input	Purpose
Selected units	Scenario being valued.
Average monthly rent	Average rent per unit before vacancy.
Other annual income	Parking, storage, laundry or commercial income.
Vacancy %	Stabilized vacancy and collection loss.

Input	Purpose
Operating expenses %	Operating expenses as a share of effective income.
Cap rate %	Capitalization rate applied to NOI.
Required profit % cost	Target development profit allowance.
Negotiation buffer	Additional margin below the calculated maximum.

Conceptual logic:
 Potential income
 - vacancy
 - operating expenses
 = NOI

NOI / cap rate
 = stabilized value

Stabilized value
 - development costs
 - required developer profit
 - negotiation buffer
 = preliminary maximum land value / offer

Decision use

Compare the seller's asking price with the preliminary maximum supported price. A negative or thin residual is a screening warning, not automatic proof that the property has no value.

7.7 CMHC / ACLP Sensitivity

Input	Meaning
LTC %	Maximum assumed loan as a percentage of eligible cost.
Interest %	Assumed annual interest rate.
Amortization years	Period used to estimate debt service.
Minimum DSCR	Minimum NOI-to-debt-service coverage ratio.

- The module sizes debt preliminarily using LTC and DSCR constraints.
- It does not confirm ACLP or MLI Select eligibility.
- It does not replace lender underwriting, CMHC scoring, premiums, recourse, borrower net-worth or completion-guarantee analysis.
- Use it to estimate an equity range and identify whether the proposed capital structure is plausible.

7.8 AI Due Diligence

1. Complete the project first

The generated package is only as useful as the entered facts and assumptions.

2. Tap Generate ChatGPT package

QRAOS assembles the property, zoning, financial and risk information.

3. Copy the package

Select and copy all generated text.

4. Open your ChatGPT conversation

Use a dedicated project conversation where possible.

5. Paste and request structured analysis

Ask for fatal risks, missing documents, questions, conditions and next actions.

6. Review the response

Do not accept facts that ChatGPT cannot verify from provided documents or current sources.

7. Paste the analysis back

Store the useful result in the QRAOS analysis field.

8. Save

Save the project after importing the response.

Recommended AI instruction

Require ChatGPT to distinguish: verified fact, user assumption, model inference and item requiring professional confirmation.

7.9 Municipal Intelligence

- Put only confirmed information in Verified municipal facts.
- Put broker claims, preliminary interpretations and unresolved items in Unverified assumptions.
- Generate the municipal question list before contacting the city.
- Record the name, title, date, department and exact wording of every municipal response.
- Prefer written confirmation or official documents over verbal guidance.

Core municipal topic	Question objective
Use and dwelling count	Confirm permitted use and legal density.
Height / storeys	Confirm both dimensional and storey limits.
Coverage / FAR	Confirm buildable massing limits.
Setbacks	Confirm site envelope and non-conformities.
Parking	Confirm ratios, exemptions and bicycle requirements.
Approval path	Identify as-of-right versus discretionary approvals.
Subdivision	Confirm minimum lot dimensions, frontage and process.
Demolition / heritage	Identify protection, delay and refusal risks.
Servicing	Confirm water, sewer and capacity constraints.
Fees / timing	Estimate process costs and schedule.

7.10 Architect AI

1. Select the best preliminary scenarios

Carry forward conservative and base cases.

2. Enter target unit mix

Specify studios, one-bedroom, two-bedroom and any accessibility targets.

3. Enter design priorities

Examples: efficient cores, family units, parking reduction, retention of existing structure.

4. Enter physical constraints

Existing building, irregular lot, slope, easements, access or adjacent conditions.

5. Generate the brief

Review the output for incorrect assumptions.

6. Send to the architect or technologist

Request a test fit, zoning matrix, risk list and next-step fee proposal.

7. Import confirmed results back into QRAOS

Replace assumptions with verified facts as the project advances.

7.11 Offer Generator

- Set the due-diligence period.
- Select only the conditions relevant to the transaction.
- Generate the draft.
- Send it to the broker, lawyer or notary for adaptation to the transaction and Québec law.
- Confirm deadlines, satisfaction standard, waiver mechanics, document access and termination rights.
- Never assume generated wording is automatically enforceable.

Condition type	Typical purpose
Development feasibility	Allows investigation of the intended project.
Zoning / approvals	Addresses permitted use and required authorizations.
Subdivision	Addresses legal creation of proposed lots.
Environmental	Allows environmental review and remediation risk assessment.
Financing / CMHC	Addresses acceptable debt and program eligibility.
Vacant possession	Addresses tenants, occupants and seller property.
Title / certificate of location	Addresses charges, servitudes, encroachments and conformity.
Inspection / engineering	Addresses building condition and professional review.

7.12 Knowledge Base

- Search for clause, due-diligence, municipal, architect or financing concepts.
- Treat starter content as a checklist, not a complete legal or technical authority.
- Developers can expand this module later using the QRAOS Master Knowledge Library.

- Every future knowledge item should include category, jurisdiction, source, date, status and review owner.

7.13 Professional Report

1. Save the project

Make sure all current changes are stored.

2. Build report

Generate the report from the active project.

3. Review every section

Check assumptions, currency, unit counts and missing information.

4. Print / Save PDF

Use the device print dialog and select Save as PDF where available.

5. Name the file consistently

Example: 2026-07-30_123-Test-St_QRAOS-Screening_v01.pdf.

6. Share with context

Mark the report Preliminary and identify items requiring verification.

8. End-to-End Real Estate Deal Workflow

Phase	QRAOS action	Output	Advance only when
1. Lead capture	Create project and enter listing facts.	Saved lead record.	Opportunity fits basic location and strategy.
2. Initial screen	Enter preliminary zoning and physical assumptions.	Initial yield and feasibility score.	No obvious fatal issue.
3. Financial screen	Run cost, residual and financing sensitivity.	Preliminary maximum price and equity range.	Economics justify further effort.
4. AI review	Generate and analyze due-diligence package.	Risk list and missing information.	Critical questions are identified.
5. Municipal inquiry	Use municipal question list and record responses.	Verified facts and approval path.	Intended use appears plausible.
6. Professional test fit	Generate architect brief and commission focused work.	Test fit and zoning matrix.	Concept works physically and legally.
7. Offer preparation	Generate draft conditions for professional review.	Conditional offer language.	Risks are covered by enforceable conditions.
8. Due diligence	Track legal, title, structural, environmental, servicing and financing work.	Decision file.	All material risks are accepted, priced or resolved.
9. Approval decision	Build report and compare against investment criteria.	Proceed, renegotiate or terminate decision.	Decision is documented and authorized.
10. Handoff	Export JSON and report for records and future system migration.	Portable project record.	Backups are retained securely.

8.1 Recommended verification hierarchy

Reliability	Example
1 - Official written evidence	Bylaw, title, certificate of location, permit, written municipal confirmation.
2 - Professional opinion	Lawyer, notary, architect, engineer, surveyor, environmental consultant or lender.
3 - Broker or seller document	Listing, declaration, invoice, warranty or correspondence.
4 - User assumption	Preliminary interpretation entered for screening.
5 - AI inference	Reasoned possibility requiring confirmation.

9. Data Backup, Export, Restore and Device Migration

9.1 Backup procedure

1. Open the correct project

Confirm project name and address.

2. Tap Save

Write current edits to browser storage.

3. Tap Export

Download the project's JSON file.

4. Rename the file

Use date, address and version.

5. Move it to a controlled folder

Examples: OneDrive, Google Drive, iCloud Drive or company document management.

6. Retain milestone versions

Keep exports at offer, end of due diligence and final decision.

Recommended naming:
YYYY-MM-DD_Address_QRAOS_Project_v01.json

Current restoration limitation

The current proof of concept may not include a user-facing JSON Import control. Preserve exported files for backup and developer migration. A future version should add validated import and merge functions.

9.2 Before replacing or resetting a device

- ☐ Export every active project.
- ☐ Save important reports as PDF.
- ☐ Confirm files open from the backup location.
- ☐ Record the deployed GitHub Pages URL.
- ☐ Do not clear the old device until the backups are verified.

10. Updating the Application

1. Back up project data

Export important projects before changing application files.

2. Obtain the new audited package

Confirm the release and change notes.

3. Extract the package

Do not upload only the ZIP.

4. Upload replacement files

Use Add file > Upload files in the same repository.

5. Commit changes

Use a clear version message.

6. Wait for GitHub Pages deployment

Check the Actions or Pages status.

7. Refresh the web version

Open the HTTPS URL in Safari or desktop browser.

8. Allow service-worker update

Close and reopen the Home Screen app if the previous version remains cached.

9. Run regression tests

Test save, calculations, prompt generation, report and export.

10. Confirm data remains

Open an existing project and verify key fields.

Developer release rule

Every release should change the service-worker cache name, include a change log and be tested at the actual GitHub project subpath.

11. Testing and Acceptance Checklist**11.1 Deployment acceptance**

- ☐ Repository is public or otherwise eligible for the selected GitHub Pages plan.
- ☐ Pages deploys from main and / (root).
- ☐ index.html is at repository root.
- ☐ Published site uses HTTPS.
- ☐ Manifest and icons load.
- ☐ Service worker registers without a fatal error.
- ☐ Application opens at the repository subpath.

11.2 Functional acceptance

- ☐ Create, save, reopen and delete/test projects as designed.
- ☐ All navigation buttons work.
- ☐ Numeric fields accept expected values.
- ☐ Feasibility calculations react to changed inputs.
- ☐ Cost and residual calculations produce plausible results.
- ☐ CMHC sensitivity responds to LTC, interest, amortization and DSCR changes.
- ☐ AI package contains current project data.
- ☐ Municipal questions and architect brief generate.
- ☐ Selected offer conditions appear in the draft.
- ☐ Knowledge search returns relevant starter content.
- ☐ Professional report builds and prints.
- ☐ Export downloads valid JSON.

11.3 Mobile acceptance

- ☐ Add to Home Screen works from Safari.
- ☐ App opens in standalone mode.
- ☐ Controls are readable without horizontal scrolling.
- ☐ Keyboard input does not block critical controls.
- ☐ Saved project remains after app restart.
- ☐ Offline launch works after the app has been opened online at least once.
- ☐ Update from an older cached version is tested.

12. Troubleshooting

Problem	Likely cause	Resolution
GitHub Pages shows 404	Deployment is incomplete, wrong branch/folder or index.html is nested.	Wait several minutes; confirm main, / root and root-level index.html.
Repository page opens instead of app	User opened github.com repository URL.	Open username.github.io/repository-name/.
Blank page	Missing files, JavaScript error or corrupted upload.	Re-upload audited files and inspect browser console.
App does not install	Not opened in Safari, HTTPS absent or manifest unavailable.	Use Safari and published HTTPS URL; confirm manifest loads.
Old version remains	Service-worker cache has not refreshed.	Open web URL, refresh, close app fully and reopen; developers should change cache version.
Project missing on another device	Data is local to each browser/device.	Use exports and a future import/cloud function.
Project disappeared	Browser website data cleared, private mode used or storage evicted.	Restore from backup through developer assistance or future import feature.
Calculations look wrong	Incorrect units, blank assumptions or inconsistent gross/net area.	Review every numeric input and assumption source.
Report omits changes	Project was not saved or report was not rebuilt.	Save, rebuild and reprint.
Export not found	Browser downloaded it to default Downloads/Files folder.	Check browser downloads and Files app.
Map does not verify zoning	Map is only an external location aid.	Use official municipal zoning sources.
ChatGPT response conflicts with project facts	AI inferred or used incomplete context.	Require evidence classification and professional confirmation.

13. Privacy, Security and Operational Controls

Public interface

Free GitHub Pages deployment normally makes the application code and built-in templates public. Do not place

passwords, API keys, confidential documents or proprietary databases in the repository.

13.1 Appropriate proof-of-concept data

- Public listing facts.
- Non-sensitive screening assumptions.
- Fictional test projects.
- General zoning questions.
- Preliminary cost and rent assumptions without confidential source documents.

13.2 Data to avoid

- Tenant identification or financial information.
- Bank details, investor records or personal guarantees.
- Private legal agreements or unredacted title documents.
- Passwords, tokens, API keys or credentials.
- Sensitive seller communications.
- Confidential appraisals or proprietary databases.

13.3 Minimum operating controls

- ☐ Export backups at defined milestones.
- ☐ Use consistent file naming and controlled storage.
- ☐ Label every report Preliminary - Subject to Verification.
- ☐ Separate verified facts from assumptions.
- ☐ Require professional review before relying on legal, zoning, engineering, environmental or financing conclusions.
- ☐ Maintain a decision log identifying who approved proceeding, renegotiating or terminating.

14. Developer Handoff and Technical Reference

14.1 Current technical stack

Layer	Current implementation
Frontend	Single static index.html containing interface, styling and JavaScript logic.
Persistence	Browser localStorage.
PWA	manifest.webmanifest plus service-worker.js.
Hosting	GitHub Pages project site under a repository subpath.
External link	OpenStreetMap search opened by the user.
AI	Manual prompt exchange with ChatGPT; no API.
Backend	None.
Build system	None.
Authentication	None.
Database	None beyond browser storage.

14.2 GitHub Pages compatibility requirements

- ☐ All internal asset paths must remain relative, such as ./manifest.webmanifest.
- ☐ Manifest id, start_url and scope must remain compatible with a project subpath.
- ☐ Service-worker registration must use a relative script path and project-level scope.
- ☐ Fetch handling must not intercept unrelated paths on the github.io origin.
- ☐ Do not require server-side routing, PHP, Python or Node runtime.
- ☐ Keep index.html at repository root when Pages publishes from / (root).
- ☐ Increment the service-worker cache name with every release.
- ☐ Retain .nojekyll.

14.3 Core functional areas developers must preserve

Area	Preserve
Project management	Create/select/save/export project behavior and active-project identity.
Data separation	Verified municipal facts versus unverified assumptions.
Calculations	Feasibility, construction cost, residual value and financing sensitivity.
Generated documents	AI package, municipal questions, architect brief, offer draft and report.
Local-first behavior	Usable without a paid API or backend for the proof-of-concept

Area	Preserve
	edition.
Mobile UX	Home Screen installation and touch-friendly navigation.
Warnings	Preliminary-use and professional-verification disclosures.

14.4 Recommended data model for production rebuild

Entity	Suggested fields
User	id, name, email, status, MFA, organization_id.
Organization	id, legal_name, settings, retention policy.
Project	id, name, address, municipality, stage, owner, timestamps.
Property facts	lot, cadastral identifiers, existing units, price, source, verification status.
Zoning fact	topic, value, unit, source, jurisdiction, effective date, verified_by, confidence.
Scenario	units, GFA, efficiency, parking, cost, rent, cap rate, outputs.
Task	category, description, owner, priority, status, due date, evidence.
Document	type, filename, storage reference, version, confidentiality, extraction status.
AI analysis	prompt version, model, source package, response, reviewer, status.
Clause	jurisdiction, category, text, source, review date, lawyer approval.
Report	template, version, generated_by, data snapshot, file reference.
Audit event	actor, action, object, timestamp, before/after hash.

14.5 Calculation governance

- Store formulas centrally instead of scattering them across UI event handlers.
- Version every formula and assumption set.
- Store units explicitly for every numeric field.
- Use decimal-safe financial arithmetic.
- Add automated tests for boundary cases, blanks, zero values and negative outputs.
- Separate deterministic calculations from AI-generated narrative.
- Display formula assumptions beside outputs.
- Support scenario cloning and sensitivity tables.

14.6 Security requirements for production

- Secure authentication with MFA.
- Role-based access and project-level permissions.
- Encrypted transport and encrypted cloud storage.
- Private object storage for documents.
- Server-side validation and audit logs.
- Backup, recovery and retention policies.
- Secrets stored outside source code.
- Canadian privacy and contractual requirements reviewed by counsel.
- Vendor and AI data-processing assessment.
- Penetration testing before confidential production use.

14.7 Developer acceptance criteria

- ☐ Existing JSON exports can be migrated without data loss.
- ☐ Every visible calculation has unit tests and documented formulas.
- ☐ Mobile, tablet and desktop layouts pass responsive testing.
- ☐ Offline or degraded-mode behavior is defined.
- ☐ AI outputs are clearly separated from verified facts.
- ☐ Every source-backed fact can retain citation metadata.
- ☐ Reports are reproducible from a saved project snapshot.
- ☐ Permissions and audit events are tested.
- ☐ Import/export format is versioned.
- ☐ Release notes and rollback instructions accompany each deployment.

15. Recommended Production Roadmap

Phase	Priority capabilities	Result
Phase 1 - Stabilize POC	Input validation, JSON import, deletion confirmation, versioned exports, improved errors, automated tests.	Reliable single-user local edition.
Phase 2 - Secure cloud core	Authentication, organization accounts, database, synchronization, backups, project permissions.	Multi-device and team use.
Phase 3 - Document intelligence	Secure upload, PDF/DOCX extraction, document classification, source citations and review workflow.	Evidence-based due diligence.
Phase 4 - Advanced underwriting	Detailed pro forma, draws, interest carry, taxes, IRR, equity waterfall, scenario sensitivities.	Investment-grade financial analysis.
Phase 5 - Municipal and CMHC knowledge	Versioned municipal rules, approval tracking, current program criteria and lender workflows.	More reliable development intelligence.
Phase 6 - Integrations	Email, calendar, cloud storage, approved	End-to-end operating system.

Phase	Priority capabilities	Result
	listing feeds and professional report automation.	

Recommended immediate next feature

Add a validated JSON Import / Restore function before relying on the proof of concept for numerous live projects. This closes the largest practical gap in the local-only workflow.

16. Quick-Reference Checklists and Glossary

16.1 Five-minute property screen

- ☐ Create project and enter address, municipality, price, existing units and lot.
- ☐ Enter only known zoning facts; label the rest as assumptions.
- ☐ Run a conservative unit scenario.
- ☐ Enter rough cost, rent, vacancy, expenses and cap rate.
- ☐ Compare asking price with residual value.
- ☐ Generate AI due-diligence package.
- ☐ Decide: reject, request more information or advance.

16.2 Before paying professionals

- ☐ No obvious prohibited use.
- ☐ No obvious lot-size, frontage, access or servicing fatal issue.
- ☐ Preliminary economics support the asking price or a realistic offer.
- ☐ Municipal questions are prepared.
- ☐ Required professional scope is narrow and defined.
- ☐ Offer conditions can preserve an exit if verification fails.

16.3 Before waiving conditions

- ☐ Title and certificate of location reviewed.
- ☐ Zoning and intended use confirmed adequately.
- ☐ Subdivision path confirmed if material.
- ☐ Inspection, engineering and environmental risks reviewed.
- ☐ Servicing and utility requirements reviewed.
- ☐ Financing and equity requirements acceptable.
- ☐ Tenancy and vacant possession evidence acceptable.
- ☐ Updated project report supports the final decision.
- ☐ Lawyer/notary confirms waiver and transaction consequences.

16.4 Glossary

Term	Meaning in this guide
PWA	Progressive Web Application installable from a supported browser.
localStorage	Browser-based local data storage tied to a website and device/browser profile.
Service worker	Browser script that caches application files and supports offline behavior.
Manifest	File describing the PWA name, icon, start URL, scope and display behavior.
FAR / FSI	Floor-area ratio: gross floor area divided by lot area.
Coverage	Building footprint as a percentage of lot area.
NOI	Net operating income after vacancy and operating expenses, before debt service.
Cap rate	NOI divided by property value.
Residual land value	Value remaining for land after development cost and required profit.
LTC	Loan-to-cost ratio.
DSCR	Debt-service coverage ratio: NOI divided by debt service.
Verified fact	Information supported by authoritative written evidence or professional confirmation.
Assumption	Unconfirmed input used for preliminary screening.
Audit trail	Record of who changed or approved information and when.

Document Control

Item	Value
Document	QRAOS Mobile PWA Master User and Developer Guide
Edition	GitHub Pages Audited Proof of Concept
Prepared	July 2026
Status	Working master guide
Primary limitation	Local browser storage; no cloud database, login or secure document repository
Required professional review	Legal, zoning, architecture, engineering, environmental, tax and financing

Final operating rule

Never waive a transaction condition or commit capital solely because the QRAOS proof of concept produces a favorable result. Use the tool to organize, calculate and question; use authoritative evidence and qualified professionals to verify.